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5 / 5 submitted

Q1. Program based on Classes and Objects

Score: 5.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;
    void displayDetails(){
        System.out.println("Roll No: " +rollNo);
        System.out.println("Name: " +name);
        System.out.println("Marks: " +marks);
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student s=new Student();
        s.rollNo=sc.nextInt();
        sc.nextLine();
        s.name=sc.nextLine();
        s.marks=sc.nextDouble();
        s.displayDetails();
        sc.close();
    }
}
```

INPUT

```
1105
Karthi
99
```

OUTPUT

```
Roll No: 1105
Name: Karthi
Marks: 99.0
```

Q2. Initialize Object Data Using Setter Methods

Score: 5.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;
    public void setRollNo(int rollNo){
        this.rollNo=rollNo;
    }
    public void setName(String name){
        this.name=name;
    }
    public void setMarks(double marks){
        this.marks=marks;
    }
    public void displayDetails(){
        System.out.print("Roll No: "+ rollNo+"\n");
        System.out.print("Name: "+ name+"\n");
        System.out.print("Marks: "+ marks);
    }

    public static void main(String args[]){
        Scanner sc=new Scanner(System.in);
        Student s=new Student();
        int rollNo=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();
        s.setRollNo(rollNo);
        s.setName(name);
        s.setMarks(marks);
        s.displayDetails();
        sc.close();
    }
}
```

INPUT

115
Mani
89

OUTPUT

Roll No: 115
Name: Mani
Marks: 89.0

Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;
    Student(){
        rollNo=101;
        name="Rahul";
        marks=85.5;
    }
    Student(int rollNo,String name,double marks){
        this.rollNo=rollNo;
        this.name=name;
        this.marks=marks;
    }
    void displayDetails(){
        System.out.println("Roll No: "+ rollNo);
        System.out.println("Name: "+ name);
        System.out.println("Marks: "+ marks);
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student s1=new Student();
        s1.displayDetails();
        int rollNo=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();
        Student s2=new Student(rollNo,name,marks);
        s2.displayDetails();
        sc.close();
    }
}
```

INPUT

```
110
mad
89.5
```

OUTPUT

```
Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 110
Name: mad
Marks: 89.5
```

Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
class Employee{
    int empId;
    String empName;
    double basicSalary;

    void getEmployeeDetails(int empId, String empName, double basicSalary){
        this.empId = empId;
        this.empName = empName;
        this.basicSalary = basicSalary;
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int id = sc.nextInt();
        sc.nextLine();
        String name = sc.nextLine();
        double salary = sc.nextDouble();

        Salary s = new Salary();
        s.getEmployeeDetails(id, name, salary);
        s.calculateSalary();
        s.displayDetails();

        sc.close();
    }
}
class Salary extends Employee {
    double hra, da, netSalary;

    void calculateSalary(){
        hra = basicSalary * 0.20;
        da = basicSalary * 0.10;
        netSalary = basicSalary + hra + da;
    }
    void displayDetails(){
        System.out.println("Employee ID: " +empId);
        System.out.println("Employee Name: " +empName);
        System.out.println("Basic Salary: " +basicSalary);
        System.out.println("HRA: " +hra);
        System.out.println("DA: " +da);
        System.out.println("Net Salary: " +netSalary);
    }
}
```

INPUT

```
101
Ramesh
500000
```

OUTPUT

```
Employee ID: 101
Employee Name: Ramesh
Basic Salary: 500000.0
HRA: 100000.0
DA: 50000.0
Net Salary: 650000.0
```

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 TC: 3/3 passed

CODE

```
import java.util.Scanner;
class Employee{
    int empId;
    String empName;
    void getEmployeeDetails(int id,String name)
    {
        empId=id;
        empName=name;
    }
    public static void main(String[] args)
    {
        Scanner sc=new Scanner(System.in);
        int id=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double salary = sc.nextDouble();
        GrossSalary g=new GrossSalary();
        g.getEmployeeDetails(id,name);
        g.getBasicSalary(salary);
        g.calculateSalary();
        g.displayDetails();
        sc.close();
    }
}
class Salary extends Employee
{
    double basicSalary;
    void getBasicSalary(double salary){
        basicSalary=salary;
    }
}
class GrossSalary extends Salary
{
    double hra,da,grossSalary;
    void calculateSalary()
    {
        hra=basicSalary*0.20;
        da=basicSalary*0.10;
        grossSalary=basicSalary+hra+da;
    }
    void displayDetails()
    {
        System.out.println("Employee ID: " + empId);
        System.out.println("Employee Name: " + empName);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("HRA: "+ hra);
        System.out.println("DA: "+ da);
        System.out.println("Gross Salary: " + grossSalary);
    }
}
```

INPUT

```
101
praveen
6092
```

OUTPUT

```
Employee ID: 101
Employee Name: praveen
Basic Salary: 6092.0
HRA: 1218.4
DA: 609.2
Gross Salary: 7919.599999999999
```